

**Load balancing:** Distributes network traffic to ensure stability and reliability.

**Self-healing:** Detects and replaces failed containers automatically.

**Auto scaling:** Provides elasticity and auto scaling. Easily scales applications up or down based on demand.

**Resource management:** Sets resource quotas to keep resources in check.

The latest version of Kubernetes is v1.31.1, released on September 11, 2024. It incorporates new features to enhance the platform's dependability, scalability, and user-friendliness.

## Industry adoption of LLMs

According to Brainy Insights, the generative AI-based LLMs market is poised for remarkable growth, with estimations pointing towards a staggering valuation of US\$ 188.62 billion by 2032. A Forbes report states that one of today's leading LLMs was trained on 1.4 trillion tokens.

As per Gartner, by 2026, generative design AI will automate 60% of the design effort for new websites and mobile apps. And by 2027, nearly 15% of new applications will be automatically generated by AI without a human in the loop. This is not happening at all today.

According to Accenture's 2023 Technology Vision report, 97% of global executives agree that foundation models will enable connections across data types, revolutionising where and how AI is used.

## Role of Kubernetes in deploying LLMs

Kubernetes is becoming increasingly popular for deploying LLMs due to its ability to handle resource-intensive applications. It allows the allocation of necessary CPU and memory resources for LLMs to run efficiently. Also, many LLMs require GPUs for processing large parameters efficiently. Kubernetes helps to distribute workloads across multiple GPUs to achieve optimal performance.

Kubernetes supports both horizontal and vertical scaling required for LLMs to handle varying workloads to ensure optimal resource utilisation and performance. Horizontal scaling adjusts the number of pods based on demand, while vertical scaling adjusts the resources allocated to each pod.

Kubernetes automates deployment, scaling, and management of LLMs, reducing the complexity of managing models. This includes automated rollouts, rollbacks, and self-healing capabilities, which are crucial for maintaining the reliability of LLMs in production. It supports cloud native architectures that allow seamless integration with cloud services and supports containerised applications for deploying LLMs.

Kubernetes allows portability -- LLMs can be easily moved between different environments such as cloud platforms and on-premises data centres with minimum

reconfiguration. And it easily integrates with AI/ML tools like LangChain and Hugging Face to deploy multiple LLMs on Kubernetes.

## Steps for deploying LLMs on Kubernetes

Several key steps must be followed to ensure efficient and scalable management of LLMs on Kubernetes.

**Set up Kubernetes cluster:** The first step is to set up the Kubernetes cluster and run it either on-premises or on cloud platforms like Amazon Elastic Kubernetes Services (EKS), Google Kubernetes Engine (GKE) and Azure Kubernetes Services (AKS).

**Access to resources:** Ensure the Kubernetes cluster has access to GPU resources either through physical GPUs or cloud based GPU instances.

**Establish container registry:** Set up container registry to store LLM Docker images. Examples of container registry are Docker Hub, Amazon Elastic Container Registry (ECR), Google Container Registry (GCR) and Azure Container Registry (ACR).

**Repository of LLM models:** Prepare a summary of pretrained LLM model files from different sources or train the required models. Containerise LLM applications using Docker or container runtime.

**Building the Docker image:** Create Dockerfile that specifies the environment and dependencies required for LLM. Use Docker to build the image from the Docker file. Push the built image to a container registry.

**Creating Kubernetes resources:** Define Kubernetes deployments to manage the pods running LLM. Set up Services to expose LLM pods to other parts of the cluster or external clients. Use ConfigMaps for configuration data and Secrets for sensitive information like API keys.

**Configuring resource requirements:** This is the critical step in deployment. The following resources need to be configured appropriately, such that the Kubernetes cluster runs efficiently and provides optimised performance.

- **Requests:** Specify the minimum resources required for LLM application. Kubernetes guarantees these resources.



Figure 1: Steps for deploying LLMs on Kubernetes